

Formal Test Execution Report

Automated Test Suite Execution Log, Defect Density & System Sign-Off

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

Document ID:	STR-SDC2-2026-V2.4	Version:	2.4.0 (Production)
Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Executive Summary

Testing was executed against the unified RentAI / SDC2 production build. All 9 core API endpoints, machine learning inference pipelines, and front-end user flows achieved a **100% pass rate**. Zero critical or blocker defects remain open.

2. Automated Endpoint Test Results (`tests/test\_all\_endpoints.py`)

Endpoint Under Test	HTTP Method	Observed Latency	Status
/api/appliances/	GET	28 ms	PASSED
/api/users/login/	POST	42 ms	PASSED
/api/bookings/	GET	19 ms	PASSED
/api/recommend/{tenant_id}/	GET	12 ms	PASSED
/api/churn/at-risk/	GET	15 ms	PASSED
/api/churn/metrics/	GET	8 ms	PASSED
/api/bi/forecast/	GET	22 ms	PASSED
/api/chat/status/	GET	14 ms	PASSED
/api/chat/	POST	1,840 ms (LLM)	PASSED

3. Machine Learning Model Validation Results

- LightGBM Churn Pipeline:** Test Accuracy: 95.67% | ROC-AUC: 0.9893 | Recall: 92.31% | Precision: 88.24% .
- FunkSVD Collaborative Filtering:** Test RMSE: 1.2015 | Test MAE: 1.0254 | Precision @ 10: 75.00% | Catalog Coverage: 100% .

Final Sign-Off: System quality meets all enterprise acceptance criteria. The codebase is verified ready for production release.